Bessel filter: Analog filter optimized for linear phase. It has almost no overshoot in the step response and similar rising and falling edges. Used to smooth time domain encoded signals.

Bidirectional filtering: Recursive method used to produce a zero phase filter. The signal is first filtered from left-to-right, then the intermediate signal is filtered from right-to-left.

Bilinear transform: Technique used to map the s-plane into the z-plane. Allows analog filters to be converted into equivalent digital filters.

Binning: Method of forming a histogram when the data (or signal) has numerous quantization levels, such as in floating point numbers.

Biquad: An analog or digital system with two poles and up to two zeros. Often cascaded to create a more sophisticated filter design.

Bit reversal sorting: Algorithm used in the FFT to achieve an interlaced decomposition of the signal. Carried out by counting in binary with the bits flipped left-for-right.

Blackman window: A smooth curve used in the design of filters and spectral analysis, calculated from: $0.42 - 0.5\cos(2\pi n/M) + 0.08\cos(4\pi n/M)$, where n runs from 0 to M.

Brightness: The overall lightness or darkness of an image. See *contrast* for comparison.

Butterfly: The basic computation used in the FFT. Changes two complex numbers into two other complex numbers.

Butterworth filter: Separates one band of frequencies from another; fastest roll-off while keeping the passband flat; can be analog or digital. Also called a *maximally flat* filter.

C: Common programming language used in science, engineering and DSP. Also comes in the more advanced *C++*.

Carrier wave: Term used in amplitude modulation of radio signals. Refers to the high frequency sine wave that is combined with a lower frequency information carrying signal.

Cascade: A combination of two or more stages where the output of one stage becomes the input for the next.

Causal signal: Any signal that has a value of zero for all negative numbered samples.

Causal system: A system that has a zero output until a nonzero value has appeared on its input (i.e., the input *causes* the output). The impulse response of a causal system is a causal signal.

Central Limit Theorem: Important theorem in statistics. In one form: a sum of many random numbers will have a Gaussian pdf, regardless of the pdf of the individual random numbers.

Cepstrum: A rearrangement of "spectrum." Used in homomorphic processing to describe the spectrum when the time and frequency domains are switched.

Charge coupled device (CCD): The light sensor in electronic cameras. Formed from a thin sheet of silicon containing a two-dimensional array of light sensitive regions called *wells*.

Chebyshev filter: Used for separating one band of frequencies from another. Achieves a faster roll-off than the Butterworth by allowing ripple in the passband. Can be analog or digital.

Chirp system: Used in radar and sonar. An impulse is converted into a longer duration signal before transmission, and compressed back into an impulse after reception.

Circular buffer: Method of data storage used in real time processing; each newly acquired sample replaces the oldest sample in memory.

Circular convolution: Aliasing that can occur in the time domain when frequency domain signals are multiplied. Each period in the time domain overflows into adjacent periods.

Circularity: The appearance that the end of a signal is connected to its beginning. This arises when considering only a single period of a periodic signal.

Classifiers: A parameter extracted from and representing a larger data set. For example: size of a region, amplitude of a peak, sharpness of an edge, etc. Used in pattern recognition.

Closing: A morphological operation defined as an erosion operation followed by a dilation operation.

Cochlea: Organ in the ear where sound in converted into a neural signal.

Cochlear nerve: Nerve that transmits audio information from the ear to the brain.

Coefficient-of-variation (CV): Common way of